

CL420 Introduction to Biochemical Engineering

Tutorial 1

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Problem 1: Concentration and reaction rate from absorbance (NADH assay)

NADH absorbs strongly at 340 nm; NAD⁺ does not. In a 1 cm cuvette a solution reads $A = 0.62$ at 340 nm, with ϵ (NADH, 340 nm) = $6220 \text{ M}^{-1}\text{cm}^{-1}$.

(a) Find [NADH].

(b) In an enzyme assay, A at 340 nm falls by 0.31 over 2.0 min. Express the reaction rate as NADH consumed per minute.

Problem 2: Cell density by OD₆₀₀, with dilution and calibration

A culture is too dense to read directly, so it is diluted 1:10 before measurement. The diluted sample reads $\text{OD}_{600} = 0.42$. For this organism a calibration gives $\text{OD}_{600} = 1.0 \Leftrightarrow 0.40 \text{ g dry cell weight per litre}$, valid only in the linear range $\text{OD} < 0.8$.

(a) Find the biomass concentration of the original (undiluted) culture.

(b) Why was the dilution necessary?

Problem 3: Metabolite quantitation with an internal standard

In LC–MS metabolomics, glucose is quantified against a stable-isotope (^{13}C) internal standard (IS) added at constant amount. The calibration is linear in the analyte-to-peak-area ratio $R = 0.020 \times C$, with C in μM . A cell-extract sample (diluted $10\times$ during preparation) gives $R = 0.75$.

Find the glucose concentration in the original extract.

Problem 4: Monod kinetics

A culture follows Monod kinetics with $\mu_{\text{max}} = 0.90 \text{ h}^{-1}$ and $K_s = 0.15 \text{ g}\cdot\text{L}^{-1}$ (glucose).

$$\mu = \mu_{\text{max}} \cdot S / (K_s + S)$$

(a) Find μ at $S = 0.05 \text{ g}\cdot\text{L}^{-1}$ (substrate nearly exhausted) and at $S = 2.0 \text{ g}\cdot\text{L}^{-1}$ (substrate plentiful).

(b) At what substrate concentration is $\mu = \frac{1}{2} \mu_{\text{max}}$? What is the physical meaning of K_s ?

Problem 5: Yield coefficient and substrate requirement

A batch fermentation starts with $20 \text{ g}\cdot\text{L}^{-1}$ glucose and a biomass yield $Y_{x/s} = 0.5 \text{ g cells}\cdot(\text{g glucose})^{-1}$. Inoculum biomass is negligible.

- (a) If all glucose is consumed, what is the final biomass concentration?
- (b) How much glucose must you supply to reach $12 \text{ g}\cdot\text{L}^{-1}$ cells?